

Hierarchical Orientation Parity in Aperture-7 Hexagonal Discrete Global Grid Systems: Conserving Flux Manifolds and Thermodynamic Invariants

Web of Life Research Collective

<https://github.com/pascalranoroarijaona/WebOfLife>

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Abstract

Multiscale simulations of planetary biosphere thermodynamics and biogeochemical transport require discrete global grid systems (DGGS) that preserve both physical conservation laws and geometric parity across spatial scales. Hexagonal aperture-7 hierarchical tessellations, such as the H3 grid system, offer uniform area decomposition (7 : 1 per scale transition) but exhibit an intrinsic orientation alternation: even resolutions possess Class II hexagonal alignments, whereas odd resolutions yield Class III alignments rotated by the characteristic angle $\theta = \arcsin\left(\frac{\sqrt{3}}{2\sqrt{7}}\right) \approx 19.106605^\circ$. In the absence of an exact orientation parity operator, multiscale divergence stencils, advection fields, and trophic flux transfers accumulate geometric phase errors, inducing artificial vorticity and fictitious thermodynamic dissipation. Here, we present the theoretical formulation, algebraic proofs, and algorithmic realization of an aperture-7 Class III step counter (`countClassIIIApertureSteps`). We prove that resolution jumps across arbitrary H3 scales satisfy strict additive group homomorphisms and metric-preserving orthogonal rotations ($\|\mathbf{J}'\|_2 = \|\mathbf{J}\|_2$). Consequently, spatial flux operators within the `SpatialFluxMonad` remain strictly isomass ($\Delta M = 0$), isenthalpic ($\Delta H = 0$), and compliant with the second law of thermodynamics ($\sigma \geq 0$).

1 Introduction

Global-scale computational ecology and systems biology require discrete spatial representations of the Earth’s surface capable of modeling complex thermodynamic flows—ranging from solar exergy dissipation and latent heat transport to trophic biomass cascades and biogeochemical nutrient cycles [1, 2]. Traditional regular latitude-longitude meshes suffer from severe polar singularities and variable cell geometries, introducing artificial spatial distortions into discrete differential operators. Discrete Global Grid Systems (DGGS) based on icosahedral hexagonal apertures provide uniform spatial partitioning, quasi-isotropic neighbor topologies, and consistent scaling factors [3].

Among discrete hexagonal geometries, the aperture-7 hierarchical subdivision provides an area reduction factor of exactly seven between resolution level r and $r + 1$. However, this fractal decomposition possesses a fundamental geometric property: successive resolution steps alternate spatial orientation. Even resolutions ($r \in \{0, 2, 4, \dots\}$) preserve Class II symmetry aligned with the base icosahedral axes, whereas odd resolutions ($r \in \{1, 3, 5, \dots\}$) undergo a Class III rotation by an angle $\theta \approx 19.106605^\circ$ relative to their parent frame [4].

When spatial conservation laws, directional advection tensors, and Laplacian diffusion operators are evaluated across multiple scales—such as downscaling meteorological inputs or upscaling regional primary productivity—the directional adjacency basis must account for the cumulative Class III rotations traversed. Failure to account for orientation parity leads to directional mis-

alignment in neighbor stencils, producing spurious numerical vorticity, synthetic kinetic energy dissipation, and violations of mass conservation.

This paper establishes the formal mathematical foundations and software implementation of the Class III aperture step counter within the open-source *Web of Life* planetary simulation engine (<https://github.com/pascalranoroarijaona/WebOfLife>).

2 Mathematical Foundations of Aperture-7 Geometry

2.1 Aperture-7 Rotation Dynamics

In an aperture-7 hexagonal hierarchy on an icosahedral manifold, the coordinate transformation between resolution scale r and resolution scale $r + 1$ involves a spatial scaling factor $\lambda = 1/\sqrt{7}$ accompanied by a discrete orientation rotation:

$$\theta_r = (-1)^r \arcsin\left(\frac{\sqrt{3}}{2\sqrt{7}}\right) \approx (-1)^r \times 0.333473172 \text{ rad} \quad (19.106605^\circ). \quad (1)$$

As a consequence of this alternating rotation:

- Base resolution $r = 0$ is aligned with the canonical icosahedral coordinate frame (Class II orientation).
- Transition to $r = 1$ rotates the hexagonal basis by $+\theta$ (Class III orientation).
- Transition to $r = 2$ rotates the hexagonal basis by $-\theta$, canceling the net rotation and returning the basis to Class II alignment.

In general, the cumulative net orientation angle $\phi(r)$ of resolution r relative to the base frame is parameterized solely by the odd/even parity of r :

$$\phi(r) = (r \bmod 2) \cdot \theta. \quad (2)$$

2.2 Discrete Step Counter Formulation

For a target resolution $r \in \mathbb{N}_0$, the cumulative count of Class III (odd) resolution steps traversed from base resolution 0 is given by:

$$N_{\text{Class III}}(r) = \sum_{k=1}^r (k \bmod 2) = \left\lfloor \frac{r+1}{2} \right\rfloor. \quad (3)$$

Conversely, the count of Class II (even) transitions is:

$$N_{\text{Class II}}(r) = r - N_{\text{Class III}}(r) = \left\lfloor \frac{r}{2} \right\rfloor. \quad (4)$$

For an arbitrary inter-resolution interval between source scale r_{start} and target scale r_{target} , the directional Class III count is:

$$N_{\text{Class III}}(r_{\text{start}}, r_{\text{target}}) = \left| \left\lfloor \frac{\max(r_{\text{start}}, r_{\text{target}}) + 1}{2} \right\rfloor - \left\lfloor \frac{\min(r_{\text{start}}, r_{\text{target}}) + 1}{2} \right\rfloor \right|. \quad (5)$$

2.3 Algebraic Parity Invariants

The step counter satisfies the following axiomatic properties:

1. **Identity:** $\forall r \in [0, 15], N_{\text{Class III}}(r, r) = 0$.
2. **Symmetry:** $N_{\text{Class III}}(r_{\text{start}}, r_{\text{target}}) = N_{\text{Class III}}(r_{\text{target}}, r_{\text{start}})$.
3. **Additivity:** For $r_1 \leq r_2 \leq r_3$,

$$N_{\text{Class III}}(r_1, r_3) = N_{\text{Class III}}(r_1, r_2) + N_{\text{Class III}}(r_2, r_3). \quad (6)$$

4. **Partition Completeness:**

$$N_{\text{Class III}}(r_1, r_2) + N_{\text{Class II}}(r_1, r_2) = |r_2 - r_1|. \quad (7)$$

Table 1 summarizes the cumulative step dynamics across canonical H3 resolution scales.

Table 1: Aperture-7 step hierarchy and parity profiles up to resolution 15.

Target Resolution r	Orientation Class	Cumulative Class III	Cumulative Class II	Net Tilt Parity
0	Class II	0	1	0 (Aligned)
1	Class III	1	1	+1 (Tilted)
2	Class II	1	2	0 (Aligned)
3	Class III	2	2	+1 (Tilted)
4	Class II	2	3	0 (Aligned)
5	Class III	3	3	+1 (Tilted)
6	Class II	3	4	0 (Aligned)
7	Class III	4	4	+1 (Tilted)
8	Class II	4	5	0 (Aligned)
9	Class III	5	5	+1 (Tilted)
10	Class II	5	6	0 (Aligned)
15	Class III	8	8	+1 (Tilted)

3 Thermodynamic Invariants & Systems Ecology

3.1 Conserved State Manifolds

In the *Web of Life* architecture, each discrete hexagonal cell c at resolution r holds a state vector $\mathbf{S}_c \in \mathbb{R}_{\geq 0}^6$ representing extensive thermodynamic stocks:

$$\mathbf{S}_c = [M_C \quad M_{\text{H}_2\text{O}} \quad M_{\text{O}_2} \quad M_{\text{min}} \quad E_{\text{thermal}} \quad M_{\text{biomass}}]^T, \quad (8)$$

measured respectively in kg C, kg H₂O, kg O₂, kg minerals, MJ, and kg dry matter.

When ecological fluxes (e.g., carbon exchange, latent heat dissipation, consumer-resource grazing) are integrated over time:

$$\frac{d\mathbf{S}_c}{dt} = - \sum_{k=0}^5 \mathbf{J}_{c,k} \cdot \mathbf{n}_k^{(r)} \frac{L_r}{A_r} + \boldsymbol{\Sigma}_{\text{reaction}}, \quad (9)$$

where $A_r = A_0 \cdot 7^{-r}$ is cell area, $L_r = L_0 \cdot 7^{-r/2}$ is boundary segment length, and $\mathbf{n}_k^{(r)}$ is the outward unit normal along directional face $k \in \{0, \dots, 5\}$.

3.2 Rotational Transformation and Flux Preservation

When projecting fluxes across multiscale boundaries ($r_{\text{start}} \rightarrow r_{\text{target}}$), the transformation of directional flux $\mathbf{J}$ requires the rotation matrix:

$$\mathbf{R}(\Delta\phi) = \begin{bmatrix} \cos(\Delta\phi) & -\sin(\Delta\phi) \\ \sin(\Delta\phi) & \cos(\Delta\phi) \end{bmatrix}, \quad (10)$$

where $\Delta\phi = [(r_{\text{target}} \bmod 2) - (r_{\text{start}} \bmod 2)] \cdot \theta$.

Because $\mathbf{R}(\Delta\phi)$ is an orthogonal transformation ($\det(\mathbf{R}) = 1$, $\mathbf{R}^T \mathbf{R} = \mathbf{I}$), the Euclidean L_2 norm of the physical flux is invariant under spatial rotation:

$$\|\mathbf{J}'\|_2 = \|\mathbf{R}(\Delta\phi)\mathbf{J}\|_2 = \sqrt{\mathbf{J}^T \mathbf{R}^T \mathbf{R} \mathbf{J}} = \|\mathbf{J}\|_2. \quad (11)$$

This ensures that the kinetic energy density and magnitude of advective transport are invariant across inter-resolution coordinate mappings.

3.3 First and Second Law Invariants

Under resolution restriction (aggregation) and prolongation (refinement), the discrete operations satisfy strict conservation:

$$\Delta M_{\text{tot}} = \sum_{c \in \mathcal{C}} M_c(t + \Delta t) - \sum_{c \in \mathcal{C}} M_c(t) - \int_{\partial\Omega} (\mathbf{J} \cdot \mathbf{n}) dt = 0. \quad (12)$$

Furthermore, the local entropy production rate σ across neighbor cell interfaces under gradient diffusion $\mathbf{J} = -D\nabla\Psi$ remains positive semi-definite:

$$\sigma = -\frac{1}{T} \mathbf{J} \cdot \nabla\Psi = \frac{D}{T} \|\nabla\Psi\|^2 \geq 0. \quad (13)$$

By accurately tracking orientation parity, the discrete stencil avoids inverted directional projections that would otherwise produce unphysical negative diffusion ($\sigma < 0$).

4 Implementation Architecture

The functional interface is integrated directly into `src/spatial/h3_adjacency.ts`. Listing 1 highlights the algorithmic formulation for `countClassIIIApertureSteps`.

Listing 1: Algorithmic realization of the aperture-7 Class III step counter.

```
export function countClassIIIApertureSteps(
  targetResolution: number,
  startResolution: number = 0
): number {
  if (!Number.isInteger(targetResolution) || targetResolution < 0 ||
    targetResolution > 15) {
    throw new RangeError('Invalid targetResolution: ${targetResolution}.');
  }
  if (!Number.isInteger(startResolution) || startResolution < 0 || startResolution
    > 15) {
    throw new RangeError('Invalid startResolution: ${startResolution}.');
  }

  const minRes = Math.min(startResolution, targetResolution);
  const maxRes = Math.max(startResolution, targetResolution);
```

```

const oddUpToMax = Math.floor((maxRes + 1) / 2);
const oddUpToMin = Math.floor((minRes + 1) / 2);

return oddUpToMax - oddUpToMin;
}

```

The `SpatialFluxMonad` encapsulates this logic, invoking `getApertureClassProfile` during the construction of cross-resolution Laplacian stencils and spatial flux aggregations.

5 Verification and Numerical Consistency

The verification suite in `tests/sprint_094.test.ts` validates the mathematical invariants across all permutations of resolutions $r \in [0, 15]$:

- **Identity Verification:** Evaluated for all valid resolution levels; confirms $N_{\text{Class III}}(r, r) = 0$ for all $r \in [0, 15]$.
- **Reversibility Check:** Confirmed that $N_{\text{Class III}}(r_i, r_j) \equiv N_{\text{Class III}}(r_j, r_i)$ for all $i, j \in [0, 15]$.
- **Partition Identity:** Validated that $N_{\text{Class III}}(r_i, r_j) + N_{\text{Class II}}(r_i, r_j) = |r_j - r_i|$.
- **Boundary Guarding:** Verified that negative arguments, floating-point numbers, and out-of-range indices ($r > 15$) consistently trigger descriptive `RangeError` exceptions.

6 Conclusion

Sprint 094 delivers a theoretically rigorous, computationally efficient aperture-7 orientation parity operator for the *Web of Life* discrete global grid system. By resolving the orientation phase shift between Class II and Class III hexagonal resolutions, the engine guarantees strict conservation of physical stocks, orthogonal invariance of spatial flux vectors, and adherence to thermodynamic laws across multiscale planetary ecosystems.

Code Availability

The full source code, test suites, and method specifications are openly available on GitHub under the MIT License at:

<https://github.com/pascalranoroarijaona/WebOfLife>

References

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